

DAY 17: STATISTICS FOR DATA SCIENCE

Complete Beginner-Friendly Study Guide | 60-Day Data Science Learning Journey

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1. What is Statistics?

Statistics is the process of collecting, organizing, analyzing, interpreting and communicating data. In Data Science, statistics helps us understand patterns, center, spread, variation and unusual observations.

2. Population vs Sample

Population is the entire group we want to study. **Sample** is a smaller group selected from the population. Data Scientists often analyze samples to learn about larger populations.

3. Numerical Data

Discrete data is countable, such as number of students. **Continuous** data is measurable and can contain decimals, such as height, weight or CGPA.

4. Mean

Mean is the average: sum of values divided by the number of values.

```
import numpy as np
marks = np.array([60, 70, 80, 90, 100])
print(np.mean(marks))
```

Output: 80.0. `np.mean()` calculates the arithmetic average.

5. Median

Median is the middle value after sorting. With an even number of values, the two middle values are averaged.

```
numbers = np.array([10, 20, 30, 40, 50])
print(np.median(numbers))

numbers = np.array([10, 20, 30, 40])
print(np.median(numbers))
```

Outputs: 30.0 and 25.0. Median is less affected by extreme values than mean.

6. Mode

```
import pandas as pd
numbers = pd.Series([10, 20, 20, 30, 40])
print(numbers.mode())
```

The mode is the most frequently occurring value. Here it is 20.

7. Range

```
marks = np.array([50, 60, 70, 80, 90])
range_value = np.max(marks) - np.min(marks)
print(range_value)
```

Maximum is 90 and minimum is 50, so range = 40.

8. Variance

```
marks = np.array([60, 70, 80, 90, 100])
print(np.var(marks))
```

Variance measures spread around the mean. NumPy gives 200.0 here using its default population formula.

9. Standard Deviation

```
marks = np.array([60, 70, 80, 90, 100])
print(np.std(marks))
```

The result is approximately 14.14. Standard deviation is the square root of variance and is easier to interpret because it uses the original scale.

10. Percentiles and Quartiles

```
marks = np.array([50, 60, 70, 80, 90, 100])
Q1 = np.percentile(marks, 25)
Q2 = np.percentile(marks, 50)
Q3 = np.percentile(marks, 75)
print(Q1, Q2, Q3)
```

Q1 is the 25th percentile, Q2 is the 50th percentile or median, and Q3 is the 75th percentile.

11. IQR

```
IQR = Q3 - Q1
print(IQR)
```

IQR means Interquartile Range. It describes the spread of the middle 50% of observations.

12. Outliers

A potential outlier can be flagged using the IQR rule: lower bound = $Q1 - 1.5 \times IQR$ and upper bound = $Q3 + 1.5 \times IQR$. A flagged value is not automatically an error. It should be investigated.

13. Pandas Statistical Functions

```
df["Marks"].mean()
df["Marks"].median()
df["Marks"].min()
df["Marks"].max()
df["Marks"].std()
df["Marks"].var()
df["Marks"].describe()
```

describe() provides count, mean, standard deviation, minimum, quartiles and maximum.

14. Complete Student Statistical Analysis Project

```
import pandas as pd
import numpy as np

data = {
    "Name": ["Ali", "Usman", "Hamza", "Ahmad", "Bilal", "Hassan", "Zain", "Omar"],
    "Marks": [85, 72, 91, 78, 65, 88, 95, 70],
    "CGPA": [3.2, 3.0, 3.6, 3.1, 2.8, 3.4, 3.8, 2.9]
}

df = pd.DataFrame(data)
marks = df["Marks"]

print("Mean:", marks.mean())
print("Median:", marks.median())
print("Mode:", marks.mode().tolist())
print("Minimum:", marks.min())
print("Maximum:", marks.max())
print("Range:", marks.max() - marks.min())
print("Variance:", marks.var())
print("Standard Deviation:", marks.std())

Q1 = marks.quantile(0.25)
Q2 = marks.quantile(0.50)
Q3 = marks.quantile(0.75)
IQR = Q3 - Q1

print("Q1:", Q1)
print("Q2:", Q2)
print("Q3:", Q3)
print("IQR:", IQR)

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(marks < lower) | (marks > upper)]
print("Lower Bound:", lower)
print("Upper Bound:", upper)
print("Potential Outliers:")
print(outliers)

print("Mean CGPA:", df["CGPA"].mean())
print("Median CGPA:", df["CGPA"].median())
print(df.describe())
```

Expected results: mean marks = 80.5, median = 81.5, minimum = 65, maximum = 95, range = 30. Pandas sample variance ≈ 118.57 and sample standard deviation ≈ 10.89 . $Q1 = 71.5$, $Q2 = 81.5$, $Q3 = 89.25$, $IQR = 17.75$. IQR bounds are 44.875 and 115.875, so there are no potential outliers by this rule. Mean CGPA = 3.225 and median CGPA = 3.15.

Important: NumPy `np.var()` and `np.std()` use population formulas by default. Pandas `Series.var()` and `Series.std()` use sample formulas by default.

15. Practice Tasks and Complete Solutions

```
# Task 1
marks = np.array([50, 60, 70, 80, 90])
print(np.mean(marks), np.median(marks), np.min(marks), np.max(marks))

# Task 2
numbers = pd.Series([10, 20, 20, 30, 40, 20])
print(numbers.mode())

# Task 3
numbers = np.array([15, 25, 35, 45, 55])
print(np.max(numbers) - np.min(numbers))

# Task 4
numbers = np.array([60, 70, 80, 90, 100])
print(np.var(numbers))
print(np.std(numbers))

# Task 5
numbers = np.array([10, 20, 30, 40, 50, 60, 70, 80])
Q1 = np.percentile(numbers, 25)
Q2 = np.percentile(numbers, 50)
```

```

Q3=np.percentile(numbers, 75)
print(Q1, Q2, Q3, Q3-Q1)

# Task 6-8
df2=pd.DataFrame({
    "Name":["Ali", "Usman", "Hamza", "Ahmad"],
    "Marks":[85, 72, 91, 78],
    "CGPA":[3.2, 3.0, 3.6, 3.1]
})
print(df2.describe())
print(df2.loc[df2["Marks"].idxmax()])
print(df2["CGPA"].mean())

# Task 9
numbers=np.array([60, 65, 70, 72, 75, 78, 300])
Q1=np.percentile(numbers, 25)
Q3=np.percentile(numbers, 75)
IQR=Q3-Q1
lower=Q1-1.5*IQR
upper=Q3+1.5*IQR
print(numbers[(numbers<lower)|(numbers>upper)])

# Task 10
# Use the complete project above.

```

16. Common Mistakes

Do not assume mean is always the best center measure. Do not treat every outlier as an error. Do not confuse sample and population standard deviation. Do not interpret standard deviation as a percentage. Do not assume correlation automatically proves causation.

17. Quick Revision

Mean = average. Median = middle. Mode = most frequent. Range = maximum - minimum. Variance = spread using squared deviations. Standard deviation = square root of variance. Q1 = 25th percentile. Q2 = median. Q3 = 75th percentile. IQR = Q3 - Q1. IQR can help flag potential outliers.

18. Final Checklist

■ Mean ■ Median ■ Mode ■ Range ■ Variance ■ Standard deviation ■ Percentiles ■ Quartiles ■ IQR ■ Outliers ■ NumPy statistics ■ Pandas describe()

Learning Path

Python → NumPy → Pandas → Data Cleaning → Data Visualization → Statistics → Machine Learning → Deep Learning